

Mathis Fajeau

PhD student, Inria Bordeaux (Flowers) | Autotelic Exploration | Evaluation of Generative Models | World Models
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RESEARCH PROFILE

PhD student in the Flowers team at Inria Bordeaux (advisor: Pierre-Yves Oudeyer), working on autotelic exploration for the automatic evaluation of large generative AI models: agents that set their own goals to discover and map a model's capabilities and vulnerabilities, instead of static benchmarks. Co-first author of Pinductor (arXiv 2605.13740, accepted at NeurIPS 2026), where an LLM proposes and refines executable POMDP world models scored under their own particle-filtered beliefs (Schölkopf group, MPI-IS / ELLIS Tübingen). Graduate of Georgia Tech (M.Sc. CS) and IMT Atlantique (M.Eng. AI & Embedded Systems); trained in theoretical mathematics (CPGE MP) and applied systems (CUDA/HIP kernels, quantization).

DOCTORAL RESEARCH

Inria Bordeaux, Flowers team

Nov 2026 to Oct 2029

PhD student | Advisor: Pierre-Yves Oudeyer | École Doctorale Mathématiques et Informatique, Université de Bordeaux Talence, France

- Thesis: *Autotelic Exploration for Automatic Evaluation of Large Generative AI Models: Adaptive Discovery and Mapping of Capabilities and Vulnerabilities*.
- Research threads: autotelic exploration, LLM-guided program induction, quality-diversity search over model behaviours.

PUBLICATION

Pinductor: Learning POMDP World Models from Observations with Language-Model Priors *arXiv 2605.13740 | NeurIPS 2026*

V. Six*, F. Panse*, M. Fajeau*, L. Da Costa*, M. Sharma, A. Amayuelas, T. Xiao, D. Hyland, P. Hennig, B. Schölkopf (* co-first). Induce executable POMDP world models from observation, action and reward trajectories alone (no privileged state supervision), by scoring LLM-proposed components under their own particle-filtered beliefs; matches state-supervised LLM baselines on five partially observable MiniGrid tasks with strictly less information. Paper: arxiv.org/abs/2605.13740 | Code: github.com/atomresearch/pinductor.

RESEARCH COLLABORATION

Max Planck Institute for Intelligent Systems (MPI-IS) / ELLIS Institute Tübingen Feb 2026 to Present

Research Collaborator | Schölkopf group | Agent Foundation Model program

Remote

- Implementing active-inference controllers via Expected Information Gain to minimize model uncertainty in interactive environments.
- Built reproducible large-scale evaluation infrastructure for active-learning baselines, with rigorous metrics and high-performance execution.

SELECTED RESEARCH & ENGINEERING PROJECTS

Autotelic Quality-Diversity for Discovering LLM Sycophancy

2026

Extension of ACES (Flowers): a search loop that invents its own multi-turn attacks against a local 7B model, scored by a judge-free logprob fitness (single constrained token, the model's cold answer as control), archived in a continuous mechanism-embedding space with distance-gated admission, and acting through five channels including forged system directives, forged assistant turns and many-shot conditioning. **164 tactics**, more diverse than ACD and ACES at matched size; matched control 16% vs 3% placebo regression; attacks transfer to an unseen 14B model. Code and report: github.com/mtsfi/autotelic-qd-sycophancy.

Full-Parameter vs LoRA Fine-Tuning of 7B LLMs

Personal project

Built a 1,536-pair instruction dataset from scratch and led fine-tuning runs of Mistral-7B end to end (Hugging Face transformers/PEFT, QLoRA, single H100); LoRA reached **72.3%** on a 1,500-question benchmark vs **46.07%** pre-trained baseline, while full-parameter fine-tuning mode-collapsed after one epoch. Code: github.com/mtsfi/fl-conversational-ai.

GRACE: Verifier-Guided Chain-of-Thought Decoding

2024 to 2025

Stepwise decoding for Mistral-7B with a contrastive-loss step verifier that checks each reasoning step against a learned criterion; **+16% on GSM8K** without additional fine-tuning.

HIP-Transformer

2024 to 2025

Decoder-only Transformer in C++/HIP with hand-written attention kernels (SRAM tiling, manual backward pass); **40x speedup** over CPU baselines, compute-bound via Roofline analysis.

PROFESSIONAL EXPERIENCE

Airbus | *Machine Learning Engineer Intern*

Feb 2025 to Aug 2025

- Architected RAG pipelines over 10 000+ engineering documents; optimized retrieval latency and context-window utilization.
- Engineered a validation protocol for LLM reasoning in safety-critical settings, benchmarked against 10+ years of ground-truth data.

Thales | *Software Engineer Intern (Deep Learning R&D)*

Apr 2024 to Jul 2024

- INT8/FP8 Transformer quantization and weight sparsity for FPGA/GPU; sub-millisecond perception-model latency via hardware-aware layer fusion and memory-access optimization.

EDUCATION

Georgia Institute of Technology

Dec 2025

M.Sc. Computer Science (Systems & Machine Learning)

Metz, France & Atlanta, USA

- GPA **3.77/4.00**. Machine Learning (A), ML for Robotics (A), Large Language Models (A), GPU Hardware (A), Big Data Systems (A), Graduate Algorithms (A).
- Teaching Assistant for MATH 2552 Differential Equations (20+ students); dynamical-systems stability and Runge-Kutta integrators.

IMT Atlantique (French Grande École, CTI accredited)

March 2026

M.Eng. AI & Embedded Systems, Heterogeneous Embedded Systems track

Brest, France

- Cumulative GPA **3.86/4.00**. Algorithms & Discrete Math (A+), Signal Processing & Automatic (A), Parallel Computing (A), Embedded HW/SW Interaction (A), High-level Circuit Design (A).

CPGE MPSI / MP, Lycée Alphonse Daudet

2020 to 2022

Theoretical Mathematics & Physics

France

- Admitted to a French Grande École via Concours Mines-Ponts. MP training in topology, analysis, linear algebra, ODEs, and Newtonian mechanics.

TECHNICAL SKILLS & INDEPENDENT STUDY

- **ML / LLMs:** PyTorch, Hugging Face transformers/PEFT; fine-tuning (SFT, LoRA/QLoRA); reinforcement learning (model-based RL, POMDPs); world models; particle filtering, belief-state estimation; quality-diversity and open-ended search; generative models, diffusion.
- **Systems / HPC:** CUDA/HIP, Triton, SRAM tiling, Roofline, Nsight/Omniperf, Slurm, Docker, CI/CD; high-performance Python and C++.
- **Online coursework:** CMU 10-714 Deep Learning Systems (Chen & Kolter, building a PyTorch-like framework from scratch); Stanford CS236 (Ermon), EE364A (Boyd); MIT 18.065 (Strang); NYU DS-GA 1008; Cornell CS4780. **Languages:** French (native), English (C1).